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27 October 1970
Superseding
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10 December 1968

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, MARK 37 MOD 0

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 37 MOD 0 impulse cartridge and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-P-223	Powder, Black
MIL-D-21625	Design and Evaluation of Cartridges and Cartridge Actuated Devices
MIL-P-46610	Primers, Percussion, Styphnate and Chlorate, for Small Arms Ammunition

FSC 1377

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS AND ASSOCIATED LISTS

Naval Air Systems Command (Code Ident 30003)

LD 419706	Cartridge, Impulse, MARK 37 MOD 0 (Formerly Code Ident 10001)
LD 537899	Container, Inner, for Cartridges, Power Source, Assembly (Formerly Code Ident 10001)
LD 537900	Container, Outer, for Cartridges, Power Source, Assembly (Formerly Code Ident 10001)
DL 311AS155	Test Fixture, Vented Bomb, MARK 37 Cartridge
2164482	Cartridge, Impulse, MARK 37 MOD 0, Assembly (Formerly Code Ident 10001)

U. S. Army Munitions Command (Code Ident 19200)

B-6200959	Primer (116M-282A)

PUBLICATIONS

Bureau of Naval Weapons (Code Ident 10001)

OP 5	Ammunition Ashore (This publication is not furnished to bidders and contractors by the procuring activity. Contractors should contact the Government Inspector for use of the publication.)

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WR-43 Preparation of Quality Assurance Provisions

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 171-178 Transportation, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should request the latest edition and supplements thereto.)

3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 37 MOD 0 impulse cartridge covered by this specification shall be manufactured in accordance with the drawings listed on LD 419706.

3.2 Materials. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referred to on the respective Naval Air Systems Command drawings unless specific approval in writing covering a departure therefrom has been obtained from the Navy cognizant field activity for cartridges prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified by the procuring activity, a preproduction sample of 55 MARK 37 MOD 0 impulse cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to starting production, is capable of producing items which comply with the technical requirements of the contract.

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3.3 Primary components. For the purposes of this specification, the primer and main propellant charge are defined as primary components. See 4.3.3.

3.4 Primer. The primers shall be furnished from lots manufactured within the preceding twelve months. The primers shall be manufactured in accordance with Drawing B-6200959 (116M-282A), loaded with Olin Industries primer mixture 282A; and shall be accepted in accordance with MIL-P-46610, Class 1. See 4.7.1.

3.4.1 X-ray. Primers shall be free of the following defects when examined by X-ray in accordance with 4.7.1.1: missing or abnormally thin charge; charge broken, cracked or porous; cracked or off-center anvil, or other misplaced or deformed inert parts; anvil abnormally close to primer cup dome; and all other observable defects in assembly.

3.5 Propellant and booster charges.

3.5.1 Quantity. All propellant charges and booster charges shall be as specified on Drawing 2164482. Charges shall be determined by weight rather than by volume measurements.

3.5.2 Black powder. The moisture content of the black powder shall be determined as specified in MIL-P-223 and at the time of loading shall be within the limits specified therein. See 4.7.2.

3.5.3 Smokeless powder, handling and loading. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.5.3.1 Smokeless powder, condition. The smokeless powder utilized shall exhibit no evidence of decomposition. See 4.7.3.

3.6 Cartridge inspection.

3.6.1 Visual inspection. Cartridges shall be free of the following visible defects: burrs, dents, deep scratches, split or cracked edges, bulged cartridge case (from crimping), sharp edges, defective sealant application around the crimped areas; and all other defects which could conceivably prevent entry into the firing chamber or adversely affect ballistic performance. See 4.7.4.

3.7 Leakage. Cartridge leakage rate shall not exceed 10^{-5} cc of air or gas per second when tested in accordance with 4.7.5.

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3.8 X-ray. Cartridges shall be free of imperfections in assembly when examined by X-ray in accordance with 4.7.6.

3.9 Ballistic requirements. Cartridges from each production lot, selected in accordance with 4.8, shall meet the requirements listed below when subjected to the test prescribed in 4.8.

3.9.1 Chamber pressure. The lower (L) limit of maximum chamber pressure shall be L = 2000 psi. See 4.8.1.2.

3.9.2 Misfire. There shall be no misfires. See 4.8.2.

3.9.3 Hangfire. There shall be no hangfires. See 4.8.3.

3.10 Change in propellant lot. Whenever a change occurs in the propellant lot, test firings shall be conducted in accordance with 4.11. No cartridge shall produce results outside the limits of 3.9.1 through 3.9.3.

3.11 Packaging, packing and marking. The packaging and packing of cartridges and the container markings shall be in accordance with Section 5. See 4.6.1.

3.12 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-414 or MIL-STD-105, whichever applies.

4.3 Lots.

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4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-414 or MIL-STD-105, whichever applies.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 55 MARK 37 MOD 0 impulse cartridges shall be prepared. Of these cartridges, 45 shall be expended in the tests listed below and ten shall be retained for investigative purposes. A preproduction sample acceptable for environmental and functional testing shall be defined as a sample which has met the requirements of paragraphs 3.3 through 3.8 of this specification. Preproduction sample acceptance shall consist of the following tests as prescribed in MIL-D-21625: 40-foot drop (3), 6-foot drop (3), 15 "g" shock (6), temperature and humidity cycling (9), vibration (6). These cartridges (with exception of the 40-foot drop cartridges), after exposure to environmental treatments, together with six untreated cartridges shall be temperature conditioned at $+70^{\circ} \pm 5^{\circ}F$ for a minimum of six hours and fired within five minutes after removal from conditioning. In addition, 12 cartridges shall be temperature conditioned for a minimum of six hours as follows: $+160^{\circ} \pm 5^{\circ}F$ (6) and $-65^{\circ} \pm 5^{\circ}F$ (6). Each of these cartridges shall be fired within three minutes after removal from conditioning. The numbers in parentheses denote the number of cartridges allocated for test. Cartridge firing shall be conducted in the test fixture of 4.8.1. All cartridges shall fire and produce results within the limits set forth in 3.9.1 and 3.9.3. The failure of any cartridge to meet the acceptance requirements as stated above shall cause rejection of the preproduction sample.

4.3.3 Production lot. Unless otherwise specified in the contract, the minimum and maximum production lot size shall be 181 and 500 cartridges, respectively. Only primary components from a single lot shall be used in a production lot of cartridges; however, one primary component production lot may be used in more than one cartridge production lot.

4.4 Gages. The contractor shall provide gages as necessary to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Classification of characteristics. The characteristics verified by the tests and examinations herein are classified as Critical, Major or Minor in accordance with WR-43. Tests and examinations that verify critical characteristics are identified by the symbol (C), and major characteristics by the symbol (M). The number following the classification symbol indicates the serial number of test or examination. Tests and examinations which are not annotated with a classification code are classified minor.

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4.6 Quality conformance inspection. Quality conformance inspection shall consist of verification of the characteristics classified on the drawings of LD 419706 and verification of the following requirements and tests:

Item	Nature of Requirement Verification	Applicable Paragraphs	Classification (WR-43)
Primary Component	Single lot	3.3, 4.3.3	(M101)
Primer	Age, loading and acceptance	3.4, 4.7.1	(C1)
Primer	X-ray	3.4.1, 4.7.1.1	(C2)
Propellant and Booster Charges	Type and weight	3.5.1	(C3)
Black Powder	Moisture content	3.5.2, 4.7.2	(C4)
Smokeless Powder	Condition	3.5.3.1, 4.7.3	(C5)
Cartridge	Visual inspection	3.6.1, 4.7.4	(M102)
Cartridge	Leakage	3.7, 4.7.5	(C6)
Cartridge	X-ray	3.8, 4.7.6	(C7)
Cartridge	Ballistic requirements	3.9, 4.8	(C8)
Cartridge	Ballistic requirements (change in propellant lot)	3.10, 4.11	(C9)
Cartridge	Packaging, packing and container markings	3.11, 4.6.1	(M103)

4.6.1 Packing, packaging and marking. Quality conformance inspection shall be conducted in accordance with the inspection criteria on the applicable drawings referenced in Section 5.

4.7 Tests.

4.7.1 Primer verification. The contractor shall provide verification that the primers used were furnished from lots which were manufactured, loaded, and accepted in accordance with the requirements of 3.4.

4.7.1.1 Primer X-ray. Examine all primers by X-ray and reject those having any of the defects of 3.4.1. Pressure-sensitive tape may be used as an aid in positioning the primers in suitable grooved racks. For the most revealing exposure, the primers are laid on their side with the anvil cutouts vertical.

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4.7.2 Moisture content of black powder. Conduct at least one test for each cartridge lot (or daily if this requires more frequent testing) and such additional tests as considered warranted by variations in conditions or process. Samples may be collected at the site of loading from the powder container, or taken from cartridges prior to sealing the black powder charge in the cartridges. The presence of moisture in excess of the limitations of MIL-P-223 shall cause suspension of loading until corrective measures have been taken.

4.7.3 Smokeless powder, condition. Smokeless powder shall be inspected in accordance with the applicable paragraphs of OP 5. Evidence of decomposition shall be cause for rejection.

4.7.4 Visual inspection. Examine all cartridges and reject those having any of the defects of 3.6.1.

4.7.5 Leakage. Each cartridge in the preproduction sample or production lot sample shall be leak tested in a dry gas or dry air leak tester. Cartridges which exhibit a leak rate in excess of 10^{-5} cc/sec of gas or air at a pressure differential of 1.0 ± 0.1 atmosphere shall be considered defective. Cartridges of the preproduction sample which exhibit a leakage defect shall be rejected. If one or more cartridges of the production lot sample exhibit a leakage defect, then a 100 percent leak test of the lot being tested shall be required at no additional expense to the government. Cartridges which exhibit a leak rate in excess of that specified above shall be rejected.

4.7.6 X-ray examination. Examine all cartridges by X-ray and any observable imperfections in assembly shall be cause for rejection.

4.8 Ballistic tests.

4.8.1 The number of cartridges selected from each production lot for the ballistic tests shall be in accordance with the requirements of Table A-2 of MIL-STD-414, Inspection Level IV and an AQL of 0.04. The cartridges shall be conditioned for a minimum of six hours at +70° ± 5°F and fired in a vented bomb test fixture manufactured in accordance with the drawings listed on DL 311AS155. Each cartridge shall be fired within five minutes after removal from conditioning. A record of pressure versus time shall be taken with suitable instrumentation. The firing chamber, orifice and orifice plate shall be thoroughly cleaned and be free of cartridge fragments, carbon and all other residue before each firing. The orifice shall be gaged before each firing and the orifice plate shall be replaced when the specified orifice diameter of .1015 + .0010 - .0000 inch changes to the extent that it is not within the limits of .1015 inch minimum to .1045 inch maximum. In addition to the sample of cartridges for the above test, 10 additional cartridges shall be furnished from each production lot for investigative purposes.

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4.8.1.1 Criteria of acceptability.

4.8.1.2 Chamber pressure. The lower (L) limit of maximum pressure shall be as specified in 3.9.1. The lot shall be acceptable if the quantity $(\overline{X} - L)/s$ is equal to or greater than the appropriate value of k, at an AQL of 0.04, when computed in accordance with Section B, Part I of MIL-STD-414.

4.8.2 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.8.3 Hangfire. One or more hangfires shall result in rejection of the lot represented. A hangfire is defined as a firing in which the elapsed time between actuation of the firing mechanism and start of pressure rise exceeds 0.050 second.

4.9 Retest. There shall be no retests.

4.10 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and that portion of the test reconducted.

4.11 Change in propellant lot. Whenever a change in propellant lot occurs, validity of the charge determination shall be verified by firing 36 cartridges which have been temperature conditioned for a minimum of six hours (12 at $-65^{\circ} \pm 5^{\circ}F$, 12 at $+70^{\circ} \pm 5^{\circ}F$ and 12 at $+160^{\circ} \pm 5^{\circ}F$). Results of these firings shall fall within the limits prescribed in 3.9.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. For definitions of packaging levels, see 6.4.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. Four cartridges shall be packaged in an inner container in accordance with the drawings listed on LD 537899.

5.1.2 Level B. Not applicable.

5.1.3 Level C. When directed by the procuring activity, those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be packed in accordance with Method III of MIL-P-116 and with the Code of Federal Regulations 49 CFR 171-178.

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5.2 Packing. For definitions of packing levels, see 6.4.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. Six inner containers as described in 5.1.1 shall be packed in accordance with the drawings listed on LD 537900.

5.2.2 Level B. Not applicable.

5.2.3 Level C. When directed by the procuring activity, cartridges intended for expenditure in preproduction sample testing and for production lot acceptance tests and are packaged in accordance with 5.1.3 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Shipping containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3 Markings.

5.3.1 Special markings. Marking of exterior containers shall be in accordance with the Code of Federal Regulations 49 CFR 171-178.

5.3.2 Normal markings. Unless otherwise specified in the contract or order, the marking information on unit packages and shipping containers shall be as specified on the drawings listed on LD 537899 and LD 537900, respectively. The marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

6.1 Intended use. The MARK 37 MOD 0 impulse cartridge was originally designed for use in the harness and foot retainer release (Propellex Chemical Corporation P/N PCC 196) of the A-5 aircraft.

6.2 Ordering information. Procurement documents should specify the following:

a. Title, number and date of this specification;

b. whether preproduction inspection is required and, if so, specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of date required for each preproduction and production lot (see 6.3).

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6.3 Data. For the information of contractors and contracting officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order.

6.3.1 Test report. Results of ballistic tests shall be forwarded to the Naval Air Systems Command (AIR-53222) and the procuring activity.

6.3.2 Data cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. For the purposes of this specification the following definitions apply.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during handling, shipment, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity for immediate use. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirement of this level.

6.5 Explosives safety precautions. Safety precautions for explosive-loaded items are contained in MIL-STD-1314.

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6.5.1 Applicability. The loading, assembly and handling of the explosives, explosive-loaded subassemblies and the finished items covered by this specification involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Air Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed in MIL-STD-1314 are mandatory. These same minimum provisions are recommended for observance by a contractor or manufacturer that explosively loads or assembles the item covered in this specification in order to fulfill his own safety responsibility.

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